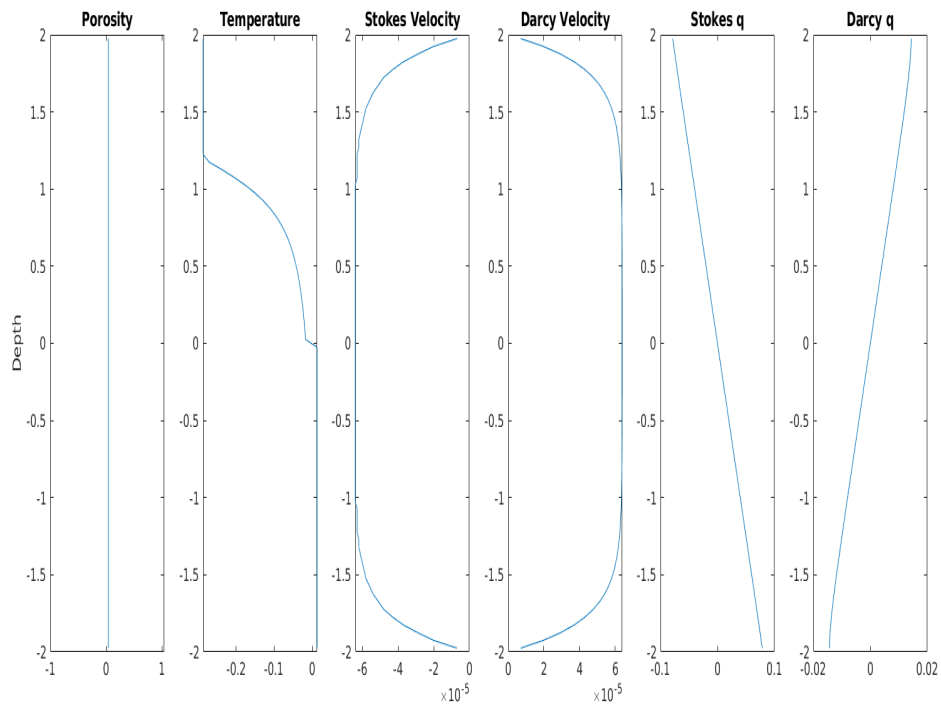
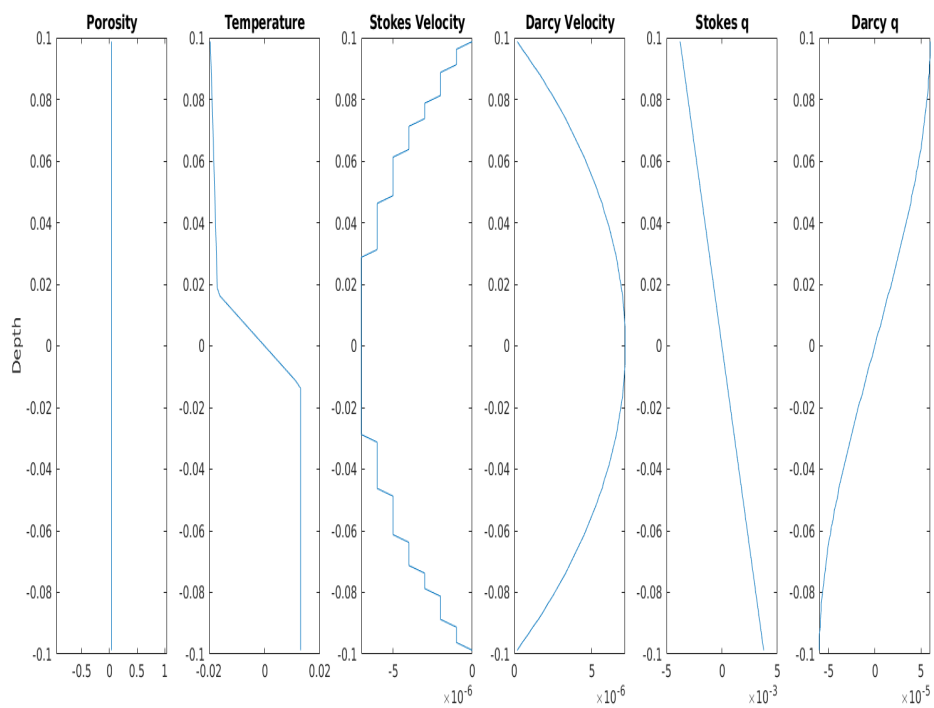
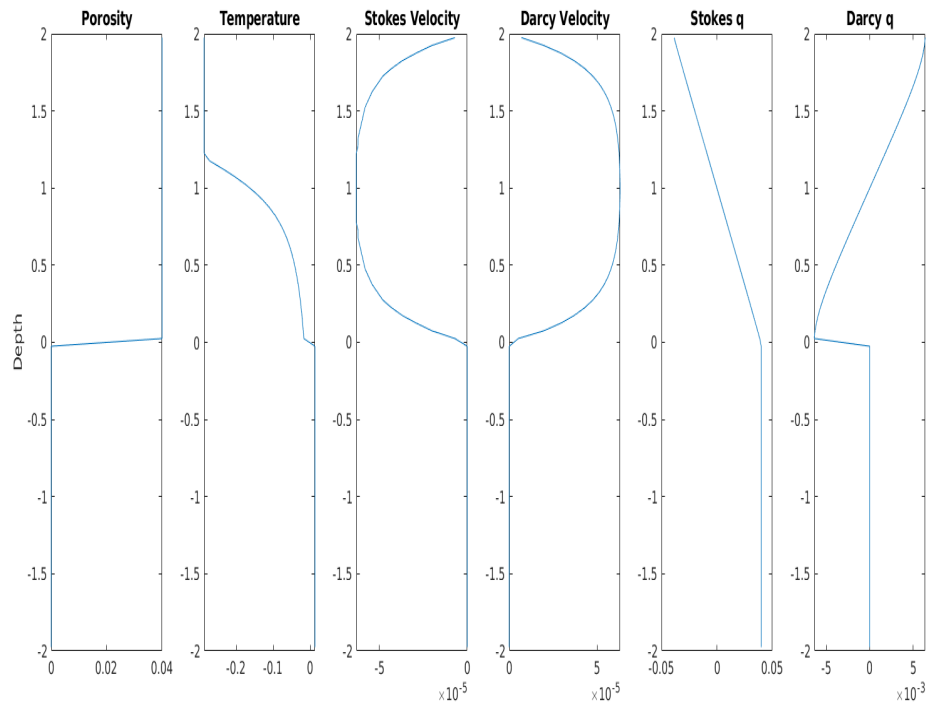
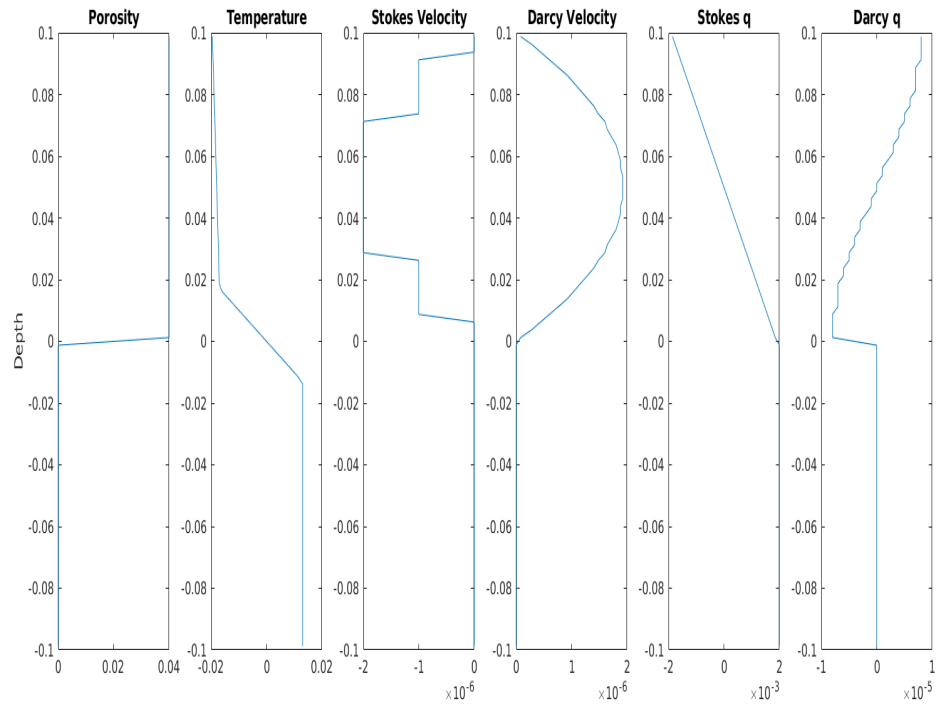
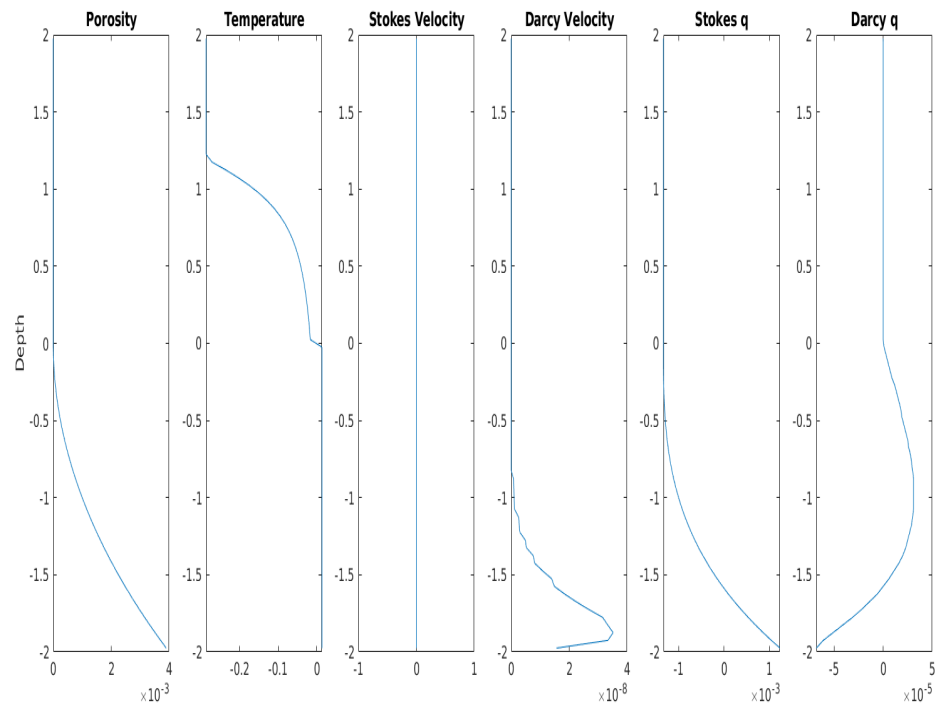


1. Outline.
 - I Introduction and Background
 - II Governing equations for partial melting problem
 - i Flow
 - ii Transport
 - iii Phase
 - III Numerics
 - i Flow (BR and Uzawa solver)
 - ii Transport (MLWENO)
 - iii Eutectic phase package
 - iv Coupled computation
 - IV Computer implementation details
 - i PETSc
 - ii Test problems
 - iii parallelization
 - V Application to real problems
 - i Mid-ocean ridge problem
 - VI Summary, conclusion and future work
2. todo.

1d plot







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